buck-5v3a — Design Document

PCB-0008-A

hwde v1 pipeline

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1 Board and Run Metadata

Field	Value
board	buck-5v3a
part number	PCB-0008-A
workspace	`boards/PCB-0008-A_buck-5v3a`
phase	P3
gate status	no gates recorded yet
state created	2026-08-08T22:10:35
state updated	2026-08-13T23:59:50

2 Overview

Brief: buck-5v3a

Build a synchronous buck converter board.

- Input 7-18 V, nominal 12 V, via 2-pin 5.08 mm screw terminal.
- Output 5 V at 3 A continuous, to a second 2-pin 5.08 mm screw terminal.
- Include reverse-polarity protection on the input.
- Ambient up to 50 C, no forced airflow.
- Board size <= 50 x 40 mm.

(Verbatim user task: "Build a synchronous buck converter board. Input 7-18V, nominal 12V, via 2-pin 5.08mm screw terminal. Output 5V at 3A continuous to a second screw terminal. Include reverse-polarity protection on the input. Ambient up to 50C, no forced airflow. <=50x40mm.")

Stackup

Chosen stackup: **'JLC04162H-7628A'** - JLCPCB impedance-controlled 4-layer, 1.6 mm nominal,

3 Requirements

Requirements: buck-5v3a

1. Function

Standalone synchronous step-down (buck) DC-DC converter board. It takes a DC input of 7-18 V (12 V nominal) on a 2-pin 5.08 mm screw terminal and produces a regulated 5 V output at 3 A continuous (15 W) on a second 2-pin 5.08 mm screw terminal, with reverse-polarity protection on the input so a swapped supply does not damage the board or the load. Power-only board: no MCU, no digital interface, no communication bus stated. Must run at up to 50 C ambient with no forced airflow, in a 50 x 40 mm outline.

2. Interfaces

- Power input (J1): 2-pin screw terminal, 5.08 mm pitch (stated). DC only. ASSUMED: through-hole terminal block, standard 5.08 mm 2-pole (e.g. Phoenix MKDS / MSTB-style clone as stocked by LCSC); rated >= 10 A and >= 250 V so it is not the limiting element at 2.5 A input. Polarity is silkscreen-marked (+ / -), and the board tolerates the reverse case by design (see Power).
- Power output (J2): 2-pin screw terminal, 5.08 mm pitch (stated). 5 V, 3 A continuous. ASSUMED: same terminal family/part as J1 for BOM consolidation, but physically separated and silk-labelled distinctly (VIN vs 5V OUT) so the two cannot be confused in the field.
- No other external interfaces stated: no USB, no RF, no header, no comms bus, no external enable/on-off line, no fault output. Indicators and test points are unspecified - see Open questions 6.

3. Power

- Source: external DC supply, 7-18 V, 12 V nominal (stated). Source TYPE is NOT stated (bench supply / AC-DC brick / battery / vehicle 12 V system) and is safety-relevant - see Open questions 1. The pipeline will not proceed on a guessed answer.
- Input current (GUESS, for sizing only): 15 W out at an assumed 90-93% efficiency -> ˜16.1-16.7 W in. At 12 V nominal ˜1.4 A; at the 7 V low-line corner ˜2.3-2.4 A. Input path (terminal, reverse-polarity device, fuse if fitted, inductor-side copper) sized for >= 2.5 A continuous.
- Output rail: single 5 V rail, 3 A continuous = 15 W (stated). No other rail stated - no auxiliary 3.3 V, no bias rail exposed. Accuracy and ripple limits are not stated - see Open questions 3.
- Topology: synchronous buck (stated) - integrated-FET regulator or controller-plus-FETs is the architect's call at P1/P2, but synchronous rectification is a requirement, not a preference (a non-synchronous design is out of spec).

- Reverse-polarity protection: REQUIRED on the input (stated). Method not stated - see Open questions 4. GUESS for thermal budgeting: a series P-channel MOSFET costs ~50-100 mW at 2.4 A, a Schottky diode ~1.0-1.2 W at the same current, which is a meaningful share of the total loss budget on a 50 C-ambient, no-airflow board.
- Loss/thermal budget (GUESS): ~1.1-1.7 W total board dissipation at 12 V in / 3 A out. With 50 C ambient, no airflow, and a 50 x 40 mm outline, copper area and via stitching under the regulator are design drivers, not afterthoughts. Peak component temperature target ASSUMED as <= 105 C junction-equivalent hotspot (derated), unless the answer to Open questions 9 imposes an enclosure that raises the internal ambient.
- No battery, no charging circuit, and no energy storage function stated on this board - PENDING confirmation that the SOURCE is not a battery (Open questions 1).

4. Environment

- Ambient: up to 50 C (stated).
- Cooling: natural convection only, no forced airflow (stated). ASSUMED: no heatsink and no thermal interface to a chassis unless the answer to Open questions 9 says the board mounts to metal.
- Minimum ambient not stated. ASSUMED: 0 C (indoor/benign). If the source turns out to be automotive or outdoor, this becomes -40 C and changes capacitor and inductor selection - covered by Open questions 1 and 9.
- Enclosure, ingress (IP) rating, vibration/shock, humidity, conformal coating: none stated - see Open questions 9.

5. Size & mounting

- Outline: 50 x 40 mm maximum (stated). Treated as a HARD cap - it binds permanently at P5 board_init and cannot be relaxed later without a restart of the mechanical baseline. Smaller is allowed; larger is not.
- Mounting holes: not stated - see Open questions 8.
- Height limit: not stated - see Open questions 8. Relevant because a 15 W buck at 50 C ambient may want a taller shielded inductor and/or a tall electrolytic input capacitor, and screw terminals themselves stand ~10-12 mm tall with wiring.
- Connector edge placement: not stated. ASSUMED: both screw terminals sit on board edges with their wire-entry openings facing outward so field wiring does not cross the board; the architect picks which edges.

6. Quantity & budget

- Build quantity: not stated - see Open questions 10. ASSUMED default of a small prototype run (qty 5) for costing purposes only.
- Target unit cost: not stated - see Open questions 10. ASSUMED: no hard per-unit cap; cost minimized by preferring JLC Basic parts and an economy fab/assembly tier, rather than hitting a fixed number.

7. Assembly

- Not stated - see Open questions 11. ASSUMED: JLCPCB PCBA, single-sided (top) SMT assembly, which suits a power board where the bottom copper is wanted as an unbroken thermal/return plane.
- The two 5.08 mm screw terminals are through-hole parts. ASSUMED: fitted by JLCPCB through-hole assembly if offered at acceptable cost for the chosen part, otherwise hand-soldered after SMT - either is acceptable and does not change the schematic or the footprint.
- No stated restriction on package sizes or on Extended/Basic library parts; ASSUMED preference for JLC Basic where a Basic part meets the electrical requirement, with Extended allowed for the regulator IC and the power inductor (Basic stock rarely covers 3 A-class buck parts).

8. Compliance/safety flags

Two flags apply; one is pending an answer.

- High current (>3 A): APPLIES at the boundary - 3 A continuous output is the stated rating, and any specified peak/inrush allowance pushes the board above 3 A (Open questions 2). Consequences carried forward: trace/copper sizing with a defined temperature rise, thermal relief and via stitching, input/output capacitor RMS ripple current ratings, connector and fuse ratings, and a defined short-circuit behavior (Open questions 5).
- >30 V: DOES NOT APPLY at the stated steady-state maximum of 18 V, but the input transient environment is unknown. If the source is a vehicle 12 V system (Open questions 1), load-dump and ISO 7637-2 transients can exceed 40 V and this flag flips to APPLIES, forcing input clamping (TVS), a higher-voltage regulator, and higher-voltage input capacitors. Not guessed - asked.
- Batteries: UNKNOWN, PENDING Open questions 1. The brief gives a voltage window (7-18 V) that is equally consistent with a bench supply, an AC-DC adapter, a 12 V lead-acid battery, or a 4S Li-ion pack. No charging function is requested on this board, but battery-source status changes reverse-polarity expectations (a mis-clipped battery is a sustained, low-impedance reverse fault, not a brief supply glitch), deep-discharge/UVLO behavior, and the minimum ambient. Per the P0 rules, this is asked, never assumed.
- Mains voltage: does not apply - the board sees DC only, and generating the DC input is out of scope for this board.
- Motors: does not apply - no motor drive stated. If the 5 V load is actually a motor or an inductive/stalling load, the peak-current answer (Open questions 2) must reflect it.
- RF transmit: does not apply - no radio function.

9. Open questions

STATUS: ALL RESOLVED 2026-08-08 by the user (Q1, Q2, Q4, Q11 answered explicitly; Q3, Q5-Q10 accepted at their stated DEFAULT). The binding answers are in section 10 - where section 10 and this section differ, section 10 wins.

1. What exactly powers the input? Pick one: (a) bench power supply, (b) AC-DC wall adapter or brick, (c) a battery - and if so, which chemistry (lead-acid, LiFePO4, Li-ion pack, other), (d) a vehicle / automotive 12 V system, or (e) something else. This is the one safety-relevant question with NO default: it decides whether the board needs load-dump/surge clamping (up to 40 V+), a wider temperature range, and battery-specific reverse-connection and under-voltage behavior. Please answer explicitly. 2. Is 3 A the absolute maximum output current, or must the board also

survive brief peaks above it (for example a motor or capacitive load drawing 4-5 A for tens of milliseconds at startup)? If there are peaks, how much current and for how long? DEFAULT: 3 A continuous is the maximum; no specified peak above it, and the regulator's own current limit provides the headroom. 3. How tight does the 5 V output need to be? Two numbers: DC accuracy (how far from exactly 5.00 V is acceptable) and ripple (how much high-frequency wobble riding on the 5 V is acceptable). DEFAULT: +/-3% DC accuracy (4.85-5.15 V) over the full line and load range, and <= 50 mV peak-to-peak output ripple. 4. Which style of reverse-polarity protection do you want? A series P-channel MOSFET wastes very little power (tens of mW) but adds a few parts; a Schottky diode is one cheap part but burns roughly 1 W as heat, which is significant on a fanless board at 50 C ambient. DEFAULT: P-channel MOSFET (low loss - the right call for this thermal budget). 5. Should the board include its own input fuse, and what should happen on an output short-circuit? Options for the fuse: a one-shot cartridge or SMD fuse (must be replaced after a fault), a resettable polyfuse (recovers by itself), or none (rely on the upstream supply's limit). DEFAULT: fit one non-resettable SMD fuse rated ˜4 A on the input, and rely on the regulator IC's built-in hiccup-mode current limit and thermal shutdown for output short-circuit and overload protection. 6. What local indicators and test access do you want? Options: a power LED on the 5 V output, test points for probing, an external enable/on-off input, a power-good signal brought out. DEFAULT: one green power LED on the 5 V output plus three test points (VIN, 5V, GND); no external enable input and no power-good output (nothing in the brief needs to talk to this board). 7. Does this board have to pass a formal EMC/emissions test (for example CISPR 32 for an information-technology product, or CISPR 25 if it goes in a vehicle), or is it an internal/prototype board where good layout practice is enough? DEFAULT: no formal EMC test campaign; design with power-loop-tight layout and normal input filtering, and leave room for an optional input filter if you later need it. 8. Mechanically: does the board need mounting holes, and is there a height limit above the PCB? If it bolts into something, what hole size and pattern? DEFAULT: four M3 holes (3.2 mm drill) inset from the corners of the 50 x 40 mm outline, and a 15 mm maximum component height (the screw terminals themselves are about 10-12 mm tall). 9. Where does the board live? Specifically: open on a bench, inside a plastic or metal enclosure, exposed to dust/water (any IP rating), subject to vibration, or needing conformal coating? Also, what is the coldest ambient it will see? DEFAULT: open board, indoors, no enclosure, no ingress rating, no vibration requirement, no conformal coating, and a 0 C minimum ambient. 10. How many boards will be built, and is there a hard per-unit cost target? DEFAULT: a prototype run of 5 boards, with no hard cost cap - cost is minimized by preferring JLC Basic parts and an economy fab tier. 11. How should it be assembled? Options: JLCPCB machine assembly of the surface-mount parts only (you hand-solder the two screw terminals), JLCPCB assembly including the through-hole screw terminals, or full hand assembly. Also: is it acceptable for all surface-mount parts to sit on the top side only? DEFAULT: JLCPCB PCBA, top-side surface-mount assembly only, with the two screw terminals added by JLCPCB through-hole assembly if the cost is reasonable and hand-soldered otherwise; bottom side kept clear as a thermal/ground plane.

10. Answers (binding, user-confirmed 2026-08-08)

A1. SOURCE: bench power supply or AC-DC adapter/brick. NOT a battery, NOT automotive. Consequences: the >30 V flag stays CLOSED - no load-dump clamping campaign, no ISO 7637-2, no -40 C rating. Input steady-state max stays 18 V; parts rated 25-40 V with a modest input TVS for hot- plug/inductive-cable ringing are sufficient. Battery flag CLOSED: no UVLO/deep-discharge behavior required, and the reverse-connect fault is supply-current-limited rather than a pack short. A2. PEAK LOAD: 3 A is the absolute maximum. No peak allowance above 3 A. Size the inductor saturation and current limit for the regulator's own limit (target ˜4-5 A), not for an external burst. A3. OUTPUT TIGHTNESS (default accepted): +/-3 % DC (4.85-5.15 V) over the full 7-18 V line and 0-3 A load range; <= 50 mV pk-pk output ripple. A4. REVERSE-POLARITY: series P-channel MOSFET in the input path (gate clamped, e.g. gate resistor + zener). Schottky rejected on thermal grounds. Target conduction loss <= 100 mW at the 2.4 A low-line input

current. A5. FUSE / SHORT-CIRCUIT (default accepted): one non-resettable SMD fuse on the input, ~4 A rating; output short-circuit and overload handled by the regulator's hiccup current limit plus thermal shutdown. A6. INDICATORS / TEST ACCESS (default accepted): one green power LED on the 5 V output; three test points (VIN, 5V, GND). No external enable input, no power-good output. A7. EMC (default accepted): no formal test campaign. Tight power-loop layout and normal input filtering; leave room for an optional input filter. A8. MECHANICAL (default accepted): four M3 (3.2 mm) holes inset from the corners; 15 mm maximum component height above the PCB. A9. ENVIRONMENT (default accepted): open board, indoors, no enclosure, no IP rating, no vibration requirement, no conformal coating, 0 C minimum ambient. 50 C maximum ambient and natural convection stand. A10. QUANTITY / COST (default accepted): prototype run of 5. No hard per-unit cap; prefer JLC Basic parts and an economy fab tier.

H1 amendments (user-ruled 2026-08-08, these SUPERSEDE A3/A5/A8 above) H1-a. FUSE: 5 A TIME-LAG, not 4 A (A5 relaxed). SMD fuses derate ~30 % at the 85-95 C board hotspot, so a 4 A part is ~2.8 A effective against 2.31 A operating - 21 % margin, thin enough to nuisance-blow. H1-b. HEIGHT: NO maximum component height. A8's 15 mm clause was an analyst default, not a stated requirement, and the board is open with no enclosure (A9). The four M3 mounting holes from A8 STAND. H1-c. TRANSIENT: A3's +/-3 % is DC-ONLY. Additionally fit a 100 uF polymer output capacitor footprint (C7) at the output; populating it is optional. H1-d. Tj <= 105 C is a SOFT design target, not a part-selection filter. The binding limit is the part's own absolute maximum.

A11. ASSEMBLY: JLCPCB PCBA, TOP-SIDE SMT ONLY. The two 5.08 mm screw terminals go to JLC through-hole assembly if reasonably priced, and are hand-soldered otherwise - either way the schematic and footprint are unchanged. Bottom side stays clear of SMT parts and is used as a thermal/return plane. Prefer Basic/Preferred parts; Extended allowed for the regulator IC and the power inductor.

4 Architecture

Decisions of record (state.json)

Phase	What	Why
P0	Input source is bench supply / AC-DC adapter; NOT battery, NOT automotive	User answer A1: closes the >30V and battery safety flags; 18V steady-state max stands, no load-dump campaign, 0C min ambient
P0	3 A is the absolute output max; no peak allowance above 3 A	User answer A2: inductor saturation and current limit sized to regulator limit (~4-5 A), not to an external burst
P0	Reverse-polarity protection = series P-channel MOSFET, target <=100 mW at 2.4 A	User answer A4: Schottky rejected on thermal grounds (50C ambient, no airflow, 50x40mm)
P0	JLCPCB PCBA top-side SMT only; screw terminals THT (JLC if priced sanely, else hand); bottom side reserved as thermal/return plane	User answer A11
P0	Defaults accepted for output tolerance (+/-3%, <=50mV ripple), 4A input fuse + hiccup limit, power LED + 3 test points, no formal EMC, 4x M3 holes + 15mm height, indoor open board 0-50C, qty 5 proto, JLC Basic preference	User accepted requirements-analyst defaults Q3,Q5-Q10 in the P0 batch

Phase	What	Why
P2	Stackup JLC04162H-7628A: 4 layers, 2 oz outer / 0.5 oz inner, 1.6 mm; outline 50x40 mm (the full cap)	check_thermal on the real part: AP63356Q 0.88 W at the 7 V corner -> Tj 95 C on 4L (pass) vs 115 C on 2L (fail). 45x35 does not fit ~557 mm2 of parts plus the GND pour and is the same JLC price tier
P2	Lead regulator AP63356QZV-7 (PWM-only sibling), not AP63357Q	PFM light-load ripple threatens the <=50 mV pk-pk spec which has no stated minimum load; light-load efficiency is not a requirement here
P2	CONFLICT RULED: datasheet theta_JA 25 C/W rejected in favour of the repo calibrated 51 C/W (4L)	Datasheet figure is a JEDEC 8710 mm2 coupon; this board gives ~2000 mm2, and 51 C/W is what the P8 verify gate applies - using the datasheet number would pass P2 and fail P8
P2	power.json's '<=90 mohm RDS(on)' part filter DELETED; loss table re-derived at 1.48 W / 91.0 % / Iin 2.35 A at 7 V	The filter was derived from generic part classes and would have rejected the entire real shortlist including the part that measurably passes
P2	ONE FLAT SCHEMATIC SHEET, no hierarchy	One function, one rail, 25 parts; also dodges the /<sheet>/<LABEL> net-prefix trap. Deviation from the default hierarchical plan, recorded per full-run.md
P2	H1: Tj <= 105 C is a soft target, not a part filter	User ruling. Computed Tj 95 C (105 C with neighbour heating) vs the part's 150 C limit is normal prototype derating; avoids a 5 A-class re-sweep the P1 data says has no stocked survivor
P2	H1: input fuse changed to 5 A TIME-LAG (was 4 A in binding answer A5)	User ruling. SMD fuses derate ~30% at the 85-95 C board hotspot: 4 A -> ~2.8 A effective vs 2.31 A operating = 21% margin, thin enough to nuisance-blow
P2	H1: 15 mm max component height DROPPED - no height limit	User ruling. A8's height clause was an analyst default, not a stated requirement; board is open with no enclosure (A9). The 4x M3 mounting holes from A8 stand. Frees inductor and cap profile choice; kills the 0.9 mm terminal margin risk
P2	H1: +/-3% is DC-only; fit a 100 uF polymer output cap footprint (C7), populate optional	User ruling. External load is unspecified; a 0->3 A step dips ~300 mV all-ceramic. ~30 mm2 and ~$0.30 of insurance that can be left unpopulated
P3	F1 stays the 32 V-rated 12H1500D (5 A time-lag); the 63 V Bel Fuse alternate at 13x cost is REJECTED	A fuse voltage rating is an arc-interruption rating, not a standoff rating. The only source this fuse can ever be asked to interrupt is the <=18 V bench/adapter supply (A1), giving nearly 2:1 margin. The ~36 V figure is a microsecond hot-plug ring that D1 (SMBJ20A) clamps and that is not a fault-interruption condition. Orchestrator ruling on the part-sourcer's OPEN 1
P3	L1 = SLO1040H6R8MTT (10.8x10 mm, 6.8 uH, DCR 22 mohm typ, Isat 10 A) over the CENKER/XR 8040 6.8 uH SKUs	The exact-size CENKER parts reach only Isat 4.55 A against the AP63356Q's 4.0 A minimum current limit - the same thin margin that disqualified 4.7 uH. H1-b dropped the height cap, which is what made the larger part admissible
P3	Q1 AO4407A Vgs(max) confirmed +/-25 V by hand from the UMW datasheet, not +/-20 V as the architecture assumed	More gate margin, not less: D3's 15 V zener now has 10 V of headroom. Recorded because the architecture's number was an assumption and downstream review will check against it

Phase	What	Why
P3	RULED: AP63356QZV-7 has 9 copper lands and NO exposed/thermal pad; '13' in V-DFN3020-13 counts TERMINALS (VIN=1-3 one land, GND=10-12 one land, SW=13, signals 4-9)	Vendor Suggested Pad Layout datasheet p.27 annotates X1(2x)+X2+X(6x)=9 and the dimension chain closes at Y3=2.825 with no room for a 10th; confirmed against Diodes standalone package-outline sheet V-DFN3020-13-SWP-Type-A1.pdf. theta_JC is plain junction-to-case, NOT theta_JC(bottom) - belly-pad evidence absent. Orchestrator hypothesis that the EasyEDA footprint was missing pads is REFUTED
P3	4 layers RETAINED, but the rationale is now the unbroken In1 GND return under the hot loop - NOT junction temperature	Recheck without the phantom exposed pad puts the 4L-vs-2L thermal gap at ~3 C, not ~20 C (the repo's 73.8 C/W 2L figure is a 1 oz calibration applied to a 2 oz board). Stackup JLC04162H-7628A unchanged; DS41948 rule 6 + topologies/buck.md s3 carry the decision instead
P3	AP63356QZV-7 STANDS: Tj 95 C at 7 V/3 A/50 C on 4L, 55 C under the 150 C recommended max	Recomputed against the real 9-land no-EP geometry. The +10 C neighbour allowance is retired as double counting - the model already brackets it. Residual risk is not thermal: DS41948 publishes no max RDS(on)/tr/tf; at 1.3x typ RDS(on) Tj is ~106 C, still 44 C clear
P3	P6 thermal prescription: 12 GND vias 0.55/0.30 through, in the pour AROUND the GND land (>=8 within 2.0 mm of the land edge, all within 4.0 mm of U1 centroid, ~1.0 mm pitch); NO via-in-pad; +VIN gets ZERO vias but >=60 mm2 contiguous F.Cu pour; >=200 mm2 F.Cu GND pour; In1/In2 unbroken within 6 mm of U1	The only heat exits are the VIN land (~1.16 mm2) and GND land (~1.16 mm2). Via-in-pad rejected on the numbers: 12 vs 16 vias is 0.4 C/W, so two in-pad vias buy ~0.2 C while putting open 0.30 mm holes in the part's only thermal AND mechanical joint; DS41948 rules 7/8 say 'around' the pins
P3	SUPERSEDED by boards/sbuck-5v3a - buck-5v3a is closed at P3 and must not be resumed	The same 7-18V -> 5V/3A synchronous-buck brief was restarted as sbuck-5v3a on 2026-08-09 with a fuller written spec (load step, ripple, efficiency floor, explicit part-selection policy); sbuck ran to P10 with a shipped fab package. buck-5v3a stopped at P3. Its P1-P3 evidence stays as reference - reports/u1-land-ruling.md (V-DFN3020-13 has 9 lands and no belly pad) and reports/thermal-recheck.md (check_thermal is an area screen) are the source of LEARNINGS 211 and 214.

Sheet plan

buck-5v3a - schematic sheet plan

1. ONE FLAT SHEET - and why that is the right answer, not laziness

No hierarchy. One root sheet, `buck-5v3a.kicad_sch`, containing all 25 placed parts. This is a deliberate deviation from the pipeline's usual hierarchical default, recorded here so it is not mistaken for an omission:

- The board has **one function** (7-18 V -> 5 V/3 A) and **one rail**. There is no second function to isolate, no repeated block to instance, no MCU/peripheral split.

- Hierarchy costs a real thing here: KiCad names a net from a hierarchical label as **‘/<sheet>/<LABEL>‘**, so every net in `constraints.json` would acquire a sheet prefix, and **a net name that does not match is silently no-op’d by every P5-P8 consumer** - the trap that cost the lumina-par run a P4 amendment. A flat sheet makes local labels come out as plain `/SW`, `/FB`, and removes that failure mode entirely.
- 25 parts fit on one A4 sheet with room to lay it out in signal-flow order (left to right: J1 -> F1 -> Q1 -> Cin/U1 -> L1 -> Cout -> J2), which is also the board floorplan. One sheet, one floorplan, one reading order.

The threshold for revisiting this: if P3/P4 adds an input filter stage or a second rail, split at that point - not before.

```
| Sheet | Blocks (‘blocks.md‘) | Refdes block | ‘#PWR‘ base |
|---|---|---|---|
| **root** (flat) | B1-B6, all of them | **1-99** per prefix | **100**
(‘#PWR0100‘+) |
```

Refdes uniqueness is structural on a single sheet. `pwr_base = 100` is still declared explicitly so that a later hierarchy split (see threshold above) starts from a convention rather than colliding with `#PWR01`.

2. Refdes assignment - BINDING ON P4

`constraints.json` (`placement.groups`, `placement.separation`, `thermal.ref`) references these refdes **by name**. A rename silently drops the constraint that mentions it - `place_anneal` collects unknown separation refs but a dropped grouping just produces a worse placement (S14 lesson). If P4 must rename, update `constraints.json` in the same commit.

```
| Ref | Part / class | Block | Notes |
|---|---|---|---|
| **J1** | 5.08 mm 2P THT screw terminal, WJ500V-5.08-2P class | B1 | left edge,
wire entry outward, silk ‘VIN 7-18V‘ |
| **F1** | 4 A 1206 one-shot fuse, 1206T4A63V class | B1 | coolest corner, next
to J1 |
| **Q1** | P-FET, AO4407A class SO-8 (AOD403 DPAK alt) | B2 | source
‘/VIN_FUSED‘, drain ‘+VIN‘ |
| **R6** | 10 kohm gate pull-down | B2 | NOT 100-330 ohm - ‘decisions.md‘ D6 |
| **D3** | zener 12-15 V, Vgs clamp | B2 | Vz <= Vgs(max) - 5 V of the chosen Q1
|
| **D2** | TVS, SMBJ20A class (20 V standoff / 32.4 V clamp) | B3 | on ‘+VIN‘,
AFTER Q1 |
| **C1, C2** | 10 uF 50 V X7R 1210 | B3 | >= 1.5 A RMS at 450 kHz; X5R
DISQUALIFIED |
| **C3** | 100 nF 50 V 0603 | B3 | at the VIN pin, same layer, shortest loop |
| **U1** | **AP63356QZV-7**, VDFN-13 3x2 | B4 | PWM-only sibling - ‘decisions.md‘
D3 |
| **C4** | 100 nF bootstrap (BST-SW) | B4 | required, not optional |
| **R1, R2** | FB divider 158k / 30.1k, **0.5 % or better** | B4 | Vout = 0.8 x
(1 + R1/R2) = 5.00 V |
| **R3, R4** | EN/UVLO divider 86.6k / 20k, 1 % | B4 | ~6.2 V rising / ~5.3 V
falling |
| **L1** | 6.8 uH, Isat >= 6 A, DCR <= 30 mohm, shielded, 125 C | B5 | **P3 must
sweep 6.8 uH - the P1 shortlist has none** |
| **C5, C6** | 22 uF 25 V X7R 1210 | B5 | ~30 uF effective after 5 V DC bias |
| **C7** | 100 uF polymer | B6 | **RESERVED refdes, NOT FITTED** (H1 open 1) |
| **J2** | same terminal family as J1 | B6 | right edge, silk ‘5V 3A‘ |
| **D1, R5** | green LED + 1 kohm | B6 | 1.9 mA |
```

```
| **TP1, TP2, TP3** | test points | B6 | '+VIN', '+5V', 'GND' |
| **H1-H4** | M3 mounting holes | - | created by 'board_init --mounting-holes 4',
board-only (not in the schematic, excluded from BOM/CPL/parity) |
```

Prefix blocks for anything P3/P4 adds: **C8+, R7+, D4+, TP4+**. Do not renumber existing refdes to close gaps.

3. Net-naming contract - BINDING ON P4

These are the **canonical names**. `constraints.json` is written against them, and a mismatch is a silent no-op in `check_current`, `check_thermal`, `rules_gen`, `planes_gen` and `route_critical`.

```
| Net | Type | From -> to | Declared in 'constraints.json' |
|---|---|---|---|
| '/VIN_RAW' | local label | J1.1 -> F1 | 'power' 3.0 A, 'pdn:  false' |
| '/VIN_FUSED' | local label | F1 -> Q1 source | 'power' 3.0 A, 'pdn:  false' |
| '+VIN' | **power symbol** | Q1 drain -> D2, C1-C3, U1.VIN, R3 | 'power' 3.0 A |
| '/VIN_GATE' | local label | Q1 gate -> R6 -> D3 | not declared (uA) |
| '/EN' | local label | R3/R4 midpoint -> U1.EN | not declared (uA) |
| '/BST' | local label | C4 -> U1.BST | not declared |
| '/SW' | local label | U1.SW -> L1, C4 low side | 'power' 3.6 A dT 20, 'pdn:
false' |
| '+5V' | **power symbol** | L1 -> C5, C6, J2, R5, R1 top | 'power' 3.6 A |
| '/FB' | local label | R1/R2 midpoint -> U1.FB | not declared |
| '/LED_A' | local label | R5 -> D1 anode | not declared |
| 'GND' | **power symbol** | everything | 'power' 3.6 A, 'plane_fed', 'pdn:
false' |
```

Rules P4 must honour:

- **Power nets are bare global power symbols** (`+VIN`, `+5V`, `GND`); everything else is a **root-sheet local label**, which on a flat sheet yields `/NAME` with no prefix. Do not use hierarchical labels - there is no hierarchy.
- **A power symbol's net name comes from its VALUE field, not its library pin name** (LEARNINGS 2026-07-28). A `+VIN` symbol cloned from `+12V` and not re-valued will produce the net `+12V` and silently orphan every `+VIN` constraint.
- `/FB` senses at the **Cout node**, not at the inductor, and is routed away from `/SW` and L1 (DS41948 Figure 47, buck.md s4).
- U1's `COMP` pin is **tied to GND** (internal compensation). `PG` is left unconnected; mark it with a no-connect flag so ERC stays clean. Pin 7 is `NC` - leave it floating, do not tie it to anything.
- `TP1` is on **'+VIN'** (post-protection), not on `/VIN_RAW`.

4. Placement groups the sheet plan implies

These become `placement.groups` in `constraints.json` and drive P6 clustering. They are listed here because the sheet layout should mirror them - a schematic drawn in this order makes the placement obvious.

1. **'hotloop'** - anchor **U1**, members **C1, C2, C3, C4**. The Cin -> VIN -> GND loop is the shortest loop on the board and outranks every other placement preference (buck.md s3, DS41948 rule 2). C4 (BST) sits directly against the BST/SW pins. 2. **'fb'** - anchor **U1**, members **R1, R2** plus **R3, R4**. FB network as close to the FB pin as possible (DS41948 rule 5); the EN divider rides along because it is the same small-signal corner of the part. 3. **'output'** - anchor **L1**, members **C5, C6**. Inductor as close to SW as possible (rule 3), output caps' GND return short (rule 4). 4.

'rpp' - anchor **Q1**, members **R6, D3, D2**. The gate network must be at the FET; the TVS lands on the same `+VIN` node.

Separations (centroid-to-centroid, `place_anneal` cost terms): **U1 <-> L1 >= 8 mm** (~2.5 mm body gap for 3x2 and 8x8 bodies) and **U1 <-> F1 >= 15 mm** (F1 in the cool corner by J1).

5. Area P6 must leave clear (no footprint, no constraint - a placement instruction)

- **~40 mm^2 beside J2 for C7**, the reserved 100 uF polymer (D6.3 x 5.8 mm class), so that a "yes" to H1 open question 1 is a BOM change and not a re-layout.
- **~60 mm^2 between J1 and F1** for the optional input filter A7 asks room for. No unpopulated footprints are added: a DNF part on a JLC PCBA BOM buys confusion, and the requirement is for *room*, not for parts.

Full architecture narrative (not inlined; see the artifact index): `architecture/blocks.md`, `architecture/decisions.md`, `architecture/power_tree.md`, `architecture/stackup.md`.

5 Schematic

Pending — produced at P4.

6 Layout

Pending — produced at P6.

7 Verification

Pending — produced at P8.

8 DFM and Fabrication

Pending — produced at P9.

9 Run Record (Appendix)

Phase digests

P0

```
# P0 Intake - digest

- buck-5v3a:  7-18 V (12 nom) -> 5 V / 3 A cont (15 W), screw terminals both
ends, RPP on input, 50 C ambient natural convection, 50x40 mm HARD cap.
- requirements.md written; check_requirements.py pass (0 violations, 1-9).
- 11 open questions batched to user, ALL resolved; binding answers = section
10.
- Safety flags CLOSED by A1 (bench/adapter source):  no load dump, no battery
case, 0-50 C. A2 caps output at 3 A, no peak allowance.  High-current
consequences (copper, cap RMS, thermal vias) carried forward.
- User calls:  P-FET RPP (not Schottky, thermal); JLC PCBA top-side SMT only,
bottom = thermal/return plane; 4 A input fuse; LED + 3 TPs; 4x M3, 15 mm
height; qty 5 proto; JLC Basic preference.
- Deviation:  requirements-analyst ran opus/high, not fable/high - Fable 5 out
of usage credits.
- Next P1:  power-architect + scout(regulator) + scout(magnetics/protection)
in parallel, then reference-design(buck stage).  No interface-spec agent.
```

P1

```
# P1 Research - digest

- Roster:  power-architect, scout(regulator), scout(powerpath),
refdesign(buck).
No interface-spec agent - no standards-bound interface on this board.
- Regulator:  AP63356QZV-7 / AP63357QZV-7 (VDFN-13 2x3, sync, 3.5 A, 3.8-32 V,
450 kHz, hiccup+OTP, ~$1.08@5).  Family is ADJUSTABLE-ONLY (catalog "Fixed"
is a scrape artifact).  5V/3A table:  L 6.8 uH, Cin 10 uF, Cout 2x22 uF,
Cbst 100 nF, FB 157k/30k.  EN self-starts, SS internal 4 ms.  refdesign
prefers the PWM-only 356Q for the 50 mV ripple spec.
- power:  4 LAYERS REQUIRED, deciding corner 7 V low line.  Eff 92.6/93.6/93.6
%
at 7/12/18 V, loss 1.03-1.20 W, Iin 2.31 A. Cin RMS ripple peaks 1.51 A at
Vin=10 V. X7R only, no electrolytics.  GND on In1+In2.
- powerpath:  PMOS AOD403/AO4407A both <100 mW at 2.4 A. TVS floor SMBJ20A
(32.4 V clamp).  Inductors single-OEM (CENKER/XR). Terminals 14.1 mm vs 15 cap.
- CONFLICT for P2:  thermal model used generic classes, never saw AP6335x
(datasheet 25 C/W vs repo calibrated 51 C/W 4L) - run check_thermal for real.
- Held for H1:  fuse 4 vs 5 A, load transient/polymer cap, Tj<=105 C hard?
```

P2

```
# P2 Architecture - digest

- 6 blocks, 25 parts, ONE FLAT SHEET (one function/one rail; also dodges the
/<sheet>/<LABEL> net-prefix trap).  Lead parts:  AP63356QZV-7, 6.8 uH
shielded
inductor, AO4407A-class P-FET, SMBJ20A TVS, fuse, WJ500V-5.08-2P terminals.
- Stackup JLC04162H-7628A: 4L, 2 oz outer / 0.5 oz inner, 1.6 mm.
Outline 50x40 (full cap - 45x35 does not fit ~557 mm2 of parts + pour).
- Thermal machine-run (check_thermal, real 74/40 mohm Rds(on)):  0.88 W at the
7 V corner -> 45.0 C rise / Tj 95 C on 4L PASS; 65.0 C / Tj 115 C on 2L FAIL.
- Conflicts ruled:  datasheet 25 C/W LOSES to repo 51 C/W (JEDEC coupon vs our
2000 mm2; 51 is what P8 applies).  power.json's <=90 mohm filter DELETED - it
rejected the whole real shortlist.  Loss retabled 1.48 W / 91.0 % at 7 V.
- Cost ~$2.60 parts/board, ~$75-85 for qty 5.  Riskiest:  2 oz outer (unpriced
fab adder; fallback to the 1 oz class is one-way safe).
- H1 APPROVED. Rulings:  Tj<=105 C soft; fuse -> 5 A time-lag (supersedes A5);
height cap DROPPED (A8 clause void, M3 holes stand); +/-3% DC-only + fit an
optional 100 uF polymer C7.
- P3 gaps to close:  6.8 uH inductor (none in any P1 sweep), 50 V X7R 10 uF,
gate zener, 0.5 % FB resistors, UMW AO4407A Vgs(max)/Rds(on) confirmation.
```

P3 *(no digest recorded)*

Gate attempt history

(no entries)

Issues

(no entries)

Human checkpoints

Id	Phase	Status	When	Note
1	P2	approved	2026-08-08T23:24:26	Architecture approved as presented: 4L JLC04162H-7628A 2oz outer, 50x40 outline, AP63356Q. Four H1 questions answered: (1) Tj<=105C is a SOFT target - proceed with AP63356Q, no re-sweep; (2) fuse relaxed to 5 A TIME-LAG (supersedes binding A5's 4 A) on SMD derating grounds; (3) 15 mm height cap DROPPED - open board, no enclosure, A8 height clause void (4x M3 holes stand); (4) +/-3% is DC-only AND fit a 100 uF polymer footprint at the output, may be left unpopulated.
2	P4	not reached		
3	P6	not reached		
4	P8	not reached		
5	P10	not reached		

Generation cost

Not recorded (no reports/cost.json).

10 Artifact Index

File	Size	Note
`state.json`	19,854 B	
`brief/brief.md`	602 B	
`requirements.md`	15,622 B	
`architecture/blocks.md`	10,148 B	
`architecture/constraints.json`	11,080 B	
`architecture/decisions.md`	18,767 B	
`architecture/power_tree.md`	10,096 B	
`architecture/sheets.md`	7,584 B	
`architecture/stackup.md`	15,643 B	
`reports/report_gen.json`	1,386 B	
`reports/thermal-recheck.md`	22,797 B	
`reports/u1-land-ruling.md`	15,930 B	